

Memory Loss

By **Mark Freedman**, MD, MSc, University of Ottawa

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Memory loss can be a symptom of [brain malfunction](#). It is one of the most common reasons that people, particularly older adults, visit a doctor. Sometimes family members notice and report the memory loss.

The biggest concern for the person, family members, and doctors is usually whether the memory loss is the first sign of [Alzheimer disease](#), a progressive and incurable form of dementia (a type of brain disorder). People with dementia have lost the ability to think clearly. Usually, if people are aware enough of their memory loss to be concerned about it, they typically do not have early dementia.

Did You Know...

- People who are aware of their memory loss typically do not have dementia.

Memories may be stored in short-term or long-term memory, depending on what they are and how important they are to the person.

- **Short-term memory** holds a small amount of information that a person needs temporarily, such as a list of things to buy at the grocery store.
- **Long-term memory**, as the name suggests, stores memories (such as the name of the person's high school) for a long time.

Short-term memory and long-term memory are stored in different parts of the brain. Long-term memory is stored in many areas of the brain. One part of the brain (the hippocampus) helps sort new information and associate it with similar information already stored in the brain. This process turns short-term memories into long-term memories. The more often short-term memories are recalled or rehearsed, the more likely they are to become long-term memories.

Causes of Memory Loss

Common causes

The most common causes of memory loss are:

- Age-related changes in memory (most common)
- Mild cognitive impairment
- Dementia
- Depression

Age-related changes in memory (called age-associated memory impairment) refer to the normal slight decline in brain function that occurs as people age. Most older adults have some memory problems. Retrieving memories of new things, such as a new neighbor's name or how to use a new computer program, takes longer. Older adults also have to rehearse new memories more often for the memories to be stored. People with this type of memory loss occasionally forget things, such as where they left their car keys. But for them, unlike people with dementia, the ability to do daily activities or to think is not impaired. Given enough time, these people usually remember, although sometimes later than is convenient. This type of memory loss is not a sign of dementia or early Alzheimer disease.

Mild cognitive impairment is an imprecise term used to describe impairments in mental function that are not severe enough to affect daily functioning. Memory loss is often the most obvious symptom. Memories are actually lost, not merely slow to be retrieved, as occurs in people with age-related changes in memory. People with mild cognitive impairment have trouble remembering recent conversations and may forget important appointments or social events, but they typically remember past events. Attention and the ability to do daily activities are not affected. People with mild cognitive impairment are at increased risk for developing dementia.

Dementia is a much more serious decline in mental function. Memory loss, particularly for recently acquired information, is often the first symptom and becomes worse with time. People who have dementia may forget entire events, not just the details. They may do the following:

- Have difficulty remembering how to do things they have done many times before and how to get to places they have often been to
- Be unable to do things that require many steps, such as following a recipe
- Forget to pay bills or keep appointments

- Forget to turn off a stove, lock the house when they leave, or take care of a child left in their care

In the early stages of dementia, people may be aware of their memory loss. But as dementia progresses, and unlike people with age-related changes in memory, they become unaware of their memory loss and often deny that they have such loss.

Finding the right word, naming objects, understanding language, and doing, planning, and organizing daily activities become more and more difficult. People with dementia eventually become disoriented, not knowing what time or even what year it is or where they are. Their personality may change. They may become more irritable, anxious, paranoid, inflexible, or disruptive.

There are many forms of dementia. [Alzheimer disease](#) is the most common. Most forms of dementia progressively worsen until the person's death.

Some conditions that increase the risk of heart and blood vessel disorders (such as high blood pressure, high levels of cholesterol, and diabetes) increase the risk of dementia.

[Depression](#) can cause a type of memory loss (called pseudodementia) that resembles memory loss due to dementia. Also, dementia commonly causes depression. Thus, determining whether dementia or depression is the cause of memory loss can be difficult. However, people with memory loss due to depression, unlike those with dementia, are aware of their memory loss and complain about it. Also, they rarely forget important current events or personal matters and usually have other symptoms, such as intense sadness, sleeping problems (too little or too much), sluggishness, or loss of appetite.

Stress can interfere with forming a memory and with recalling a memory, partly by causing preoccupation and thus preventing people from paying attention.

Less common causes

Many disorders can cause a deterioration of mental function that resembles dementia.

Some of these disorders may be reversed with treatment. They include the following:

- [Hypothyroidism](#) (an underactive thyroid gland)
- [Normal pressure hydrocephalus](#) (due to excess fluid around the brain)
- [Subdural hematoma](#) (pocket of blood under the outer layer of the membranes covering the brain)

- [Vitamin B12 deficiency](#)

Other disorders are only partially reversible. How much they can be reversed depends on how much tissue has been damaged. They include:

- Disorders that interfere with the supply of blood to the brain, such as a cardiac arrest and certain types of [stroke](#)
- Unusually long seizure
- [Head injuries](#)
- [Brain infections](#)
- [HIV \(human immunodeficiency virus\) infection](#)
- [Brain tumors](#)
- Overuse of certain medications or substances (including alcohol)

In people with these disorders, treatment can sometimes improve memory and mental function. If damage is more extensive, treatment may not improve mental function but can often prevent further deterioration.

In [delirium](#), memory is affected, but memory loss is not the most noticeable symptom. Rather, people with delirium are very confused, disoriented, and incoherent. Severe alcohol withdrawal (delirium tremens), a severe bloodstream infection (sepsis), lack of oxygen (as may result from pneumonia), and many other disorders can cause delirium, as can use of illicit drugs.

Evaluation of Memory Loss

When evaluating memory loss, doctors first determine whether the cause is [delirium](#) or another reversible cause. Reversible causes require immediate treatment.

Doctors then focus on determining whether the cause of memory loss is normal age-related changes in the brain, mild cognitive impairment, depression, or early dementia.

Warning signs

In people with memory loss, certain symptoms are cause for concern:

- Difficulty doing usual daily activities
- Difficulty paying attention and fluctuations in level of consciousness—symptoms that suggest delirium

- Symptoms of depression (such as loss of appetite, suicidal thoughts, difficulty sleeping, and slowing of speech and general activity)

When to see a doctor

People with warning signs should see a doctor. They should see a doctor immediately if they:

- Cannot pay attention and seem very confused, unfocused, and disoriented—symptoms that suggest delirium
- Feel depressed and are thinking of hurting themselves
- Have other symptoms that suggest a problem with the nervous system, such as headaches, difficulty using or understanding language, sluggishness, vision problems, or dizziness

People who do not have warning signs but are concerned about their memory or have difficulty doing basic daily activities should call their doctor. The doctor can determine how quickly they need to be seen based on other symptoms they have and the severity of the symptoms.

What the doctor does

Doctors ask about the person's symptoms and [medical history](#). Doctors then do a physical examination. Having a family member present is helpful because people with memory difficulties may not be able to describe their symptoms accurately. What doctors find during the history and physical examination often suggests a cause and the tests that may need to be done (see table

[Some Causes and Features of Memory Loss](#)).

Doctors often talk to the person and the person's family members separately because family members may not feel free to describe the symptoms candidly with the person listening.

Doctors ask specific questions about the memory loss:

- What types of things the person forgets (for example, whether the person forgets words or names or gets lost)
- When the memory problems started
- Whether memory loss is getting worse
- How the memory loss is affecting the person's ability to function at work and at home

Doctors also ask whether the person has other symptoms, such as difficulty using or understanding language and changes in their eating and sleeping habits or mood.

They ask about all disorders the person has had and all the medications, illicit drugs, over-the counter medications, and nutritional supplements the person is taking to check for possible causes. Doctors also ask the person about any unusual dietary habits. Information about the person's education, jobs, and social activities can help doctors better assess the person's previous mental function and gauge the severity of the problem. Doctors ask whether any family members have had dementia or early mild cognitive impairment.

During the physical examination, doctors evaluate all body systems but focus on the nervous system ([neurologic examination](#)), including evaluation of mental function (mental status testing).

In [mental status testing](#), doctors ask people to answer questions or do specific tasks to evaluate various aspects of mental function, such as:

- Orientation to time, place, and person: State the current date and place and who they are.
- Attention: Repeat a short list of words.
- Concentration: Spell "world" backwards or repeat their phone number forward, then backward.
- Short-term memory: Recall the short list of words after several minutes.
- Long-term memory: Answer questions about the distant past.
- Use of language: Name common objects and body parts, and read, write, and repeat certain phrases.
- Ability to understand spatial relationships: Copy simple and complex structures (for example, using building blocks) and draw an object such as clock, cube, or house.

This testing also assesses abstract thinking, comprehension, the ability to follow commands and solve math problems, awareness of the illness, and mood.



Some Causes and Features of Memory Loss

Cause	Common Features*	Diagnostic Approach†
Age-related memory changes (age-associated memory impairment)	Occasional forgetfulness of such things as names or the location of car keys No effect on thinking, other mental functions, or the ability to do daily activities	A doctor's examination alone (particularly a neurologic examination and mental status testing to assess functions such as attention, orientation, and memory)
Mild cognitive impairment	Memory loss that is more severe than expected for a person's age, particularly difficulty remembering recent events and conversations (short-term memory loss) No effect on the ability to do daily activities An increased risk of developing dementia	A doctor's examination (particularly a neurologic examination and mental status testing to assess functions such as attention, orientation, and memory) Sometimes formal neuropsychologic testing, which resembles mental status testing but evaluates function in more detail
Dementia	Memory loss that becomes worse as time passes, eventually with no awareness of the loss Difficulty using and understanding	A doctor's examination (particularly a neurologic examination and mental status testing to assess functions

